Lecture 24 — Linear dynamical systems

Mon Oct 19

Last updated: October 8, 2026.

Boyd video (optional): Lecture 26 — linear dynamical systems · ENGR108 playlist (Boyd)

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Reading: Boyd 9.1–9.3. Related objectives: Lectures. Schedule.

Bring paper. Warm-up while attendance is taken, then fill-in notes, board examples, and **Your turn**.

Learning objectives

By the end of class, students should be able to:

1. Represent an evolving system as $x_{t+1} = Ax_t$.
2. Interpret the state vector and transition matrix.
3. Build a transition matrix from a flow description.
4. Compute and interpret the next few states.

Warm-up

From Lecture 19, the matrix

$$A = \begin{bmatrix} 0.9 & 0.2 \\ 0.1 & 0.8 \end{bmatrix}$$

sends (urban, rural) population fractions today to tomorrow. If today's state is $x_0 = (0.6, 0.4)$, compute $x_1 = Ax_0$. In one sentence: what do the two entries of x_1 mean?

Fill in

A **linear dynamical system** (discrete time) has the form

$$x_{t+1} = _____,$$

where $t = 0, 1, 2, \ldots$ is the time index.

- The **state** x_t is an n-vector that completely describes the system at time t.
- The matrix A is the **dynamics / transition** matrix. Its size is _____ (same n as the state).

Interpretation of columns. The jth column of A is the next state if the current state is _____ (one unit in compartment j, nothing elsewhere).

A few steps by hand.

$$x_1 = Ax_0, \qquad x_2 = Ax_1 = A(Ax_0), \qquad x_3 = Ax_2 = \cdots.$$

(We will package "t steps at once" as A^t later; today focus on one step at a time.)

Building A from a flow story. For each compartment j, ask: if everything starts in j, where does it go next? Those destinations become column j.

Worked in class

W1. State and matrix. A campus tracks (on-campus residents, off-campus) each week as a 2-vector x_t. Someone proposes

$$A = \begin{bmatrix} 0.7 & 0.15 \\ 0.3 & 0.85 \end{bmatrix}.$$

(a) What does entry A_{12} mean in words? (b) What does the first column of A represent? (c) If $x_0 = (800, 200)$, compute x_1 and interpret both entries (people, not fractions).

W2. Build A from flows. Two towns, Arcata and Eureka, exchange residents each year:

- Of those in Arcata, 80% stay and 20% move to Eureka.
- Of those in Eureka, 90% stay and 10% move to Arcata.

Order the state as $x_t = (\text{Arcata}, \text{Eureka})$. Write the 2×2 transition matrix A.

W3. Next few states. With A from W2 and $x_0 = (5000, 7000)$, compute x_1 and x_2 by hand. Does the Arcata population rise or fall over these two years?

W4. Epidemic sketch (Boyd-style). Let $x_t = (S_t, I_t)$ be (susceptible, infected) counts among a fixed population of size $N = S_t + I_t$. A simple model uses

$$A = \begin{bmatrix} 0.95 & 0.10 \\ 0.05 & 0.90 \end{bmatrix}.$$

(a) Check that each column sums to 1. Why is that natural here? (b) If $x_0 = (190, 10)$, compute x_1. Did infections go up or down?

Your turn

Y1. True or false, with a one-line reason: if $x_{t+1} = Ax_t$, then knowing x_t and A determines x_{t+1} uniquely.

Y2. Three age groups (youth, adult, senior) with transition

$$A = \begin{bmatrix} 0.6 & 0 & 0 \\ 0.4 & 0.7 & 0 \\ 0 & 0.3 & 0.8 \end{bmatrix}.$$

Interpret A_{21} and A_{33} in one phrase each. If $x_0 = (100, 200, 50)$, compute x_1.

Y3. Build A. Two warehouse inventory levels $x_t = (\mathrm{A}, \mathrm{B})$. Each day:

- Warehouse A keeps 70% of its stock and ships 30% to B.
- Warehouse B keeps 60% and ships 40% to A.

Write A. Then with $x_0 = (100, 50)$, find x_1.

Y4. With $A = \begin{bmatrix} 1.1 & 0 \\ 0 & 0.5 \end{bmatrix}$ and $x_0 = (2, 8)$, compute x_1 and x_2. Which component is growing? Which is shrinking?

Answers (Your turn)

Y1. **True.** The rule $x_{t+1} = Ax_t$ is a function of the current state: one matrix–vector product gives the unique next state.

Y2. $A_{21} = 0.4$: fraction (or count factor) of youth that become adults in one step. $A_{33} = 0.8$: fraction of seniors that remain seniors.

$$x_1 = Ax_0 = \begin{bmatrix} 60 \\ 180 \\ 100 \end{bmatrix}.$$

Y3.

$$A = \begin{bmatrix} 0.7 & 0.4 \\ 0.3 & 0.6 \end{bmatrix}, \qquad x_1 = \begin{bmatrix} 90 \\ 60 \end{bmatrix}.$$

(Column 1: keep 70% in A, ship 30% to B. Column 2: ship 40% of B to A, keep 60% in B.)

Y4. $x_1 = (2.2, 4)$, $x_2 = (2.42, 2)$. First component grows (factor 1.1 each step); second shrinks (factor 0.5).